# IGCSE 0607 Practice Questions — E2.4 — Combined Questions & Markscheme

## Indices II

**Total Questions:** 21 — **Total Marks:** 113 Marks

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- **E2.4** Indices II
  → **Paper 2 (Non-Calculator)** *(17 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*


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**1** **(a)** Simplify.

$$4x^2 \times 3x^{-5}$$

..... **[1]**

**(b)** Simplify.

$$\left(\frac{8x^9}{27}\right)^{\frac{1}{3}}$$

..... **[2]**

**2** **(a)** Simplify.

$$4x^3 \times 5x^{-2}$$

..... **[1]**

**(b)** Simplify.

$$\left(\frac{2x^4}{y^3}\right)^2$$

..... **[2]**

**(c)** Solve the equation.

$$3^p = \frac{1}{27}$$

$p =$ ..... **[1]**

**3** **(a)** Simplify.

$w^4 \times w^7$

..... **[1]**

**(b)** Simplify.

$(3x^3)^2$

..... **[2]**

**(c)** Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

**4** **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

**(b)** Simplify.

$(3q^3)^2$

..... **[2]**

**(c)** Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

**5** **(a)** Work out the exact value of

$$32^{-\frac{2}{5}}$$

..... **[2]**

**(b)** Solve the equation.

$$5^{x+2} = 125^{x}$$

$x =$ ..... **[3]**

**6** **(a)** Solve the equation.

$16^x = 2$

$x =$ ..... **[1]**

**(b)** Work out the exact value of

$125^{\frac{2}{3}} \times 5^{-2}$

..... **[2]**

**(c)** Solve the equation.

$5^{2x-1} = 125^x$

$x =$ ..... **[3]**

**7** **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

**(b)** Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

**(c)** Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

**8** **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

**9** **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{81x^{12}}{16y^{8}}\right)^{-\frac{3}{4}}$$

..... **[3]**

**10** **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

**(b)** Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

**11** **(a)** Simplify fully.

$$\frac{24x^{\frac{3}{2}}y^{-2}}{6x^{-\frac{1}{2}}y}$$

..... **[2]**

**(b)** Solve the equation.

$$9^x \times 27^{x-1} = 3\sqrt{3}$$

$x =$ ..... **[4]**

**12** **(a)** Simplify fully.

$$\frac{12x^{\frac{3}{2}}y^{-2}}{4x^{-\frac{1}{2}}y^{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$4^{2x-1} \times 8^{x+1} = 32^{x}$$

$x =$ ..... **[4]**

**13** **(a)** Find the value of $x$ when

$$2^x = \frac{1}{8}$$

$x =$ ..... **[1]**

**(b)** Simplify

$$(4y^6)^{\frac{1}{2}}$$

..... **[2]**

**(c)** Work out the exact value of

$$16^{-\frac{3}{4}}$$

..... **[1]**

**14** **(a)** Simplify fully

$$\frac{12x^5y^3}{3x^2y^{-1}}$$

..... **[2]**

**(b)** Solve the equation

$$27^{x-1} = 9^{2x+1}$$

$x =$ ..... **[3]**

**15** **(a)** Simplify

$$\left(\frac{8x^9}{27y^6}\right)^{-\frac{2}{3}}$$

..... **[3]**

**(b)** Find the values of $a$ and $b$ given that

$$\frac{5^{a-2} \times 25^b}{125^{a+1}} = 5^4$$

$a =$ ..... , $b =$ ..... **[4]**

**16** **(a)** Simplify.

$5x^4 \times 3x^{-2}$

..... **[2]**

**(b)** Find the value of

$16^{\frac{3}{4}}$

..... **[2]**

**17** **(a)** Simplify fully.

$$\left(\frac{2x^5}{3y^2}\right)^3$$

..... **[3]**

**(b)** Solve the equation.

$$9^x = 27^{x+1}$$

$x =$ ..... **[3]**

**18** **(a)** Solve the equation.

$$4^x \times 8^{2x-1} = 16$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\frac{12x^{\frac{3}{2}} \times 2x^{-\frac{1}{2}}}{8x^{-3}}$$

..... **[3]**

**19** **(a)** Simplify.

$y^5 \times y^{-2}$

..... **[1]**

**(b)** Work out the value of

$5^{-2}$

..... **[1]**

**(c)** Simplify.

$(2x^4)^3$

..... **[2]**

**20** **(a)** Work out the exact value of

$$125^{\frac{2}{3}}$$

..... **[2]**

**(b)** Simplify fully.

$$\frac{15x^4y^{-3}}{5x^{-2}y^5}$$

..... **[3]**

**21** **(a)** Solve the equation.

$$9^{2x} = 27^{x+1}$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{16a^8}{81b^{12}}\right)^{-\frac{3}{4}}$$

..... **[3]**

**1** **(a)** $12x^{-3}$ or $\frac{12}{x^3}$ **[B1]**

**(b)** $\frac{2x^3}{3}$ or $\frac{2}{3}x^3$ **[B2]**

Correct cube root of coefficient $\frac{2}{3}$ OR correct index $x^3$ seen **[M1]**

[Back to Question] *[Total Marks: 3]*

**2** **(a)** $20x$ **[B1]**

**(b)** Numerator $4x^8$ or denominator $y^6$ seen **[M1]**

$\frac{4x^8}{y^6}$ **[A1]**

**(c)** $-3$ **[B1]**

[Back to Question] *[Total Marks: 4]*

**3** **(a)** $w^{11}$ **[B1]**

**(b)** $3^2$ or 9 seen, or $x^6$ seen **[M1]**

$9x^6$ **[A1]**

**(c)** $\frac{15}{5} = 3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**4** **(a)** $p^{11}$ **[B1]**

**(b)** $9q^6$ **[B2]**

Partial correct answer $9q^k$ or $cq^6$ **[B1]**

**(c)** $(\sqrt[3]{27})^2$ or $3^2$ or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

[Back to Question] *[Total Marks: 5]*

**5** **(a)** $\frac{1}{4}$ or 0.25 **[B2]**

Correct intermediate step e.g., $2^{-2}$ or $\left(\frac{1}{32}\right)^{\frac{2}{5}}$ or $32^{\frac{1}{5}} = 2$ **[M1]**

**(b)** Writes 125 as $5^3$ **[M1]**

Equates indices: $x + 2 = 3x$ **[M1]**

1 **[A1]**

[Back to Question] *[Total Marks: 5]*

**6** **(a)** $\frac{1}{4}$ or 0.25 **[B1]**

**(b)** 25 or $\frac{1}{25}$ seen **[M1]**

1 **[A1]**

**(c)** $125 = 5^3$ seen or implied **[B1]**

$2x - 1 = 3x$ **[M1]**

$x = -1$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**7** **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

**(b)** $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

**(c)** $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\frac{2a^2}{b}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**8** **(a)** 8 **[B1]**

$x^9$ **[B1]**

$y^6$ as denominator or $y^{-6}$ **[B1]**

**(b)** $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**9** **(a)** Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

**(b)** Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**10** **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any $k$ or $cx^{\frac{7}{6}}$ for any $c$ **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

**(b)** Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or $3^{5x-1}$ **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**11** **(a)** $4x^2$ or $y^{-3}$ or $\frac{1}{y^3}$ seen **[M1]**

$\frac{4x^2}{y^3}$ or $4x^2y^{-3}$ **[A1]**

**(b)** $3^{2x}$ or $3^{3(x-1)}$ seen **[B1]**

$3\sqrt{3} = 3^{1.5}$ or $3^{\frac{3}{2}}$ seen **[B1]**

$2x + 3x - 3 = 1.5$ or equivalent equation in $x$ **[M1]**

$x = 0.9$ or $\frac{9}{10}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**12** **(a)** $3x^2y^{-6}$ or $\frac{3x^2}{y^6}$ **[B3]**

Coefficient 3 seen **[M1]**

Correct subtraction of indices $x^{\frac{3}{2}-\left(-\frac{1}{2}\right)}$ leading to $x^2$ OR $y^{-2-4}$ leading to $y^{-6}$ **[M1]**

**(b)** Expresses all terms as powers of 2: $(2^2)^{2x-1} \times (2^3)^{x+1} = (2^5)^x$ **[M1]**

Correctly adds powers on the LHS: $2(2x-1) + 3(x+1)$ or $4x - 2 + 3x + 3$ **[M1]**

Forms linear equation equating LHS and RHS powers: $7x + 1 = 5x$ **[M1]**

$-\frac{1}{2}$ or $-0.5$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**13** **(a)** $-3$ **[B1]**

**(b)** $2y^3$ **[B2]**

If not B2, then B1 for 2 or $y^3$ seen **[B1]**

**(c)** $\frac{1}{8}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

**14** **(a)** $4x^3y^4$ **[B2]**

If not B2, then B1 for two of 4, $x^3$, $y^4$ correct **[B1]**

**(b)** $(3^3)^{x-1} = (3^2)^{2x+1}$ or $3^{3(x-1)} = 3^{2(2x+1)}$ **[M1]**

$3x - 3 = 4x + 2$ **[M1]**

$-5$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**15** **(a)** $\left(\frac{27y^6}{8x^9}\right)^{\frac{2}{3}}$ or $\frac{4x^6}{9y^4}$ (inverted or squared/cubed root correctly applied to some terms) **[M1]**

$\left(\frac{3y^2}{2x^3}\right)^2$ **[M1]**

$\frac{9y^4}{4x^6}$ or $2.25y^4x^{-6}$ **[A1]**

**(b)** $\frac{5^{a-2} \times 5^{2b}}{5^{3(a+1)}} = 5^4$ (expressing all terms as powers of 5) **[M1]**

$a - 2 + 2b - 3(a + 1) = 4$ **[M1]**

$-2a + 2b - 5 = 4$ or $-2a + 2b = 9$ **[A1]**

Accept any pair of values satisfying the equation, e.g. $a = 0, b = 4.5$ or $a = -4.5, b = 0$ or $b = a + 4.5$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**16** **(a)** 15 or $x^2$ in a product **[M1]**

$15x^2$ **[A1]**

**(b)** $\sqrt[4]{16}$ or $2^3$ or $4096^{\frac{1}{4}}$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 4]*

**17** **(a)** Numerator $8x^{15}$ **[B1]**

Denominator $27y^6$ **[B1]**

$\frac{8x^{15}}{27y^6}$ **[A1]**

**(b)** Writing both sides with a common base of 3, e.g. $3^{2x} = 3^{3(x+1)}$ **[M1]**

$2x = 3x + 3$ **[M1]**

$-3$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**18 (a)** Expressing all terms with a base of 2, e.g. $2^{2x} \times 2^{3(2x-1)} = 2^4$ **[M1]**

Correct linear equation for the exponents: $2x + 6x - 3 = 4$ or $8x = 7$ **[M1]**

$\frac{7}{8}$ or 0.875 **[A1]**

**(b)** Simplifying the numerator to $24x^1$ (or $24x$) **[M1]**

Correct manipulation of the coefficients $\frac{24}{8} = 3$ or indices $x^{1-(-3)}$ **[M1]**

$3x^4$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**19 (a)** $y^3$ **[B1]**

**(b)** $\frac{1}{25}$ or 0.04 **[B1]**

**(c)** $8x^{12}$ **[B2]**

If B0, allow [M1] for $8x^k$ or $kx^{12}$ where $k \neq 0$

**[Back to Question]** *[Total Marks: 4]*

**20 (a)** $(\sqrt[3]{125})^2$ or $5^2$ or $125^2 = 15625$ and $\sqrt[3]{15625}$ **[M1]**

25 **[A1]**

**(b)** 3 **[B1]**

$x^6$ **[B1]**

$y^{-8}$ or $\frac{1}{y^8}$ **[B1]**

Note: Award [B3] for a fully correct final answer of $\frac{3x^6}{y^8}$ or $3x^6y^{-8}$

**[Back to Question]** *[Total Marks: 5]*

**21 (a)** $(3^2)^{2x} = (3^3)^{x+1}$ or $3^{4x} = 3^{3x+3}$ **[M1]**

$4x = 3x + 3$ **[M1]**

3 **[A1]**

**(b)** Inverts fraction to remove negative power: $\left(\frac{81b^{12}}{16a^8}\right)^{\frac{3}{4}}$ **[M1]**

Applies power $\frac{1}{4}$ (fourth root) to all terms: $\frac{3b^3}{2a^2}$ **[M1]**

Cubes the result to get final answer: $\frac{27b^9}{8a^6}$ or $3.375b^9a^{-6}$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*